MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG1DSCECT100 - Emerging Electronics

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E)*

Part A

Multiple Choice Questions. Answer **All** Questions
Each question carries 1 mark

1.	The temperature coefficient of resistance is (a) positive for semiconductors and insulators (b) negative for conductors and semiconductors (c) negative for semiconductors and insulators (d) none	U	1
2.	The pentavalent impurities like antimony and phosphorus added to intrinsic semiconductors are called (a) acceptor or P-type impurities (b) donor or P-type impurities (c) acceptor or N-type impurities (d) donor or N-type impurities	U	1
3.	What is the consequence of exceeding a diode's PIV? (a) Improved conductivity (b) Diode breakdown (c) Increased efficiency (d) No effect	U	1
4.	The direction of conventional current flow in an NPN transistor is (a) From emitter to collector (b) From base to collector (c) From collector to emitter (d) From emitter to base	U	2
5.	Which voltage increases the channel size in JFET? (a) Positive Vgs (b) Negative Vgs (c) Positive Vds (d) Negative Vds	U	2
6.	How can you improve the output of a half-wave rectifier? (a) Increase frequency (b) Use a capacitor (c) Increase voltage (d) Add more diodes	A	2
7.	Where would you typically find an LDR? (a) In microphones (b) In light-sensitive circuits (c) In temperature sensors (d) In sound systems	U	3
8.	Which switch type is best for simple on/off control of a single device? (a) SPST (b) DPST (c) DPDT (d) SPDT	K	3
9.	What is the role of the coil in a mechanical relay? (a) Generates heat (b) Generates motion	U	3

	(c) Creates a spark (d) Closes contacts		
10	What type of protection does an MCB provide? (a) Over-voltage protection (b) Thermal protection (c) Short-circuit protection (d) Both thermal and short-circuit protection	U	3

1*10 = 10

Part B

Short Answer Questions. Answer **4** Questions
Each question carries 5 marks

11.	Compare Direct Current versus Alternating Current.	U	1
12.	Find the equivalent capacitance when two 10 μF capacitors connected in series and parallel.	A	1
13	Draw and explain the V-I characteristics of PN junction diode	U	1
14	Based in their complexity explain the classification of IC.	U	2
15	What is a Ripple factor? What is its value of a Half Wave and Full Wave Rectifier?	U	2
16	How does a Miniature Circuit Breaker (MCB) protect against short circuits?	U	3

4 * 5 = 20

Part C

Essay Questions. Answer **2** Questions
Each question carries 10 marks

17	Explain the working principle of a Zener diode as a voltage regulator, including its symbol and key components of the circuit.	U	1
18	Compare and Contrast NPN and PNP transistor. Explain the working of NPN Transistor	U	2
19	Describe the working of a center-tapped full-wave rectifier, including the role of each component and its advantages over a half-wave rectifier.	A	2
20	What is an LDR? Describe the applications of Light Dependent Resistors (LDRs) in real-world devices and systems.	K	3

2 * 10 = 20